

UX-0010 — Butlerly Low-Fidelity Wireframe Specification

Version 1.0

Purpose

This document defines the required low-fidelity wireframe set for Butlerly Version 1 across smartphone, tablet, and desktop. It translates the approved experience blueprint, journeys, information architecture, user flows, screen inventory, AI behavior, design foundations, cross-device continuity, and accessibility standard into implementation-ready screen structures and interaction annotations.

1. Wireframe Objective

The wireframes must validate structure, hierarchy, task flow, states, and platform adaptation before visual styling begins. They are not polished mockups. They must make product behavior testable without relying on color, illustration, branding, or animation.

The wireframe set must:

- Cover every P0 screen and state in UX-0005.
- Represent smartphone first, then tablet and desktop adaptations.
- Preserve one shared product model while using native platform patterns.
- Include normal, empty, loading, offline, degraded, error, success, and recovery states.
- Show AI evidence, confidence, correction, consent, and explanation.
- Treat cross-device synchronization, device trust, pairing, merge, and revocation as deferred future-design topics, not P0 MVP requirements.
- Be usable for prototype construction, usability testing, engineering estimation, and acceptance criteria.

2. Fidelity Rules

Low-fidelity wireframes use grayscale structure only.

Allowed:

- Boxes, lines, labels, icons represented by simple placeholders, and realistic interface copy.
- Relative hierarchy and approximate spacing.
- Component and screen IDs.
- Interaction arrows, state notes, accessibility notes, and platform annotations.
- Representative data that is clearly fictional.

Not allowed:

- Final brand colors.
- Decorative imagery.
- Final illustration style.
- Unapproved visual effects.
- Ambiguous lorem ipsum where real wording affects comprehension.
- Hidden behavior that is not annotated.

3. Frame Naming and Traceability

Every frame uses:

WF-[platform]-[screen ID]-[state]-[sequence]

Platforms:

- PH — smartphone.
- TB — tablet.
- DT — desktop.

Examples:

- WF-PH-HOME-001-DEFAULT-01
- WF-TB-RECORD-DETAIL-DEFAULT-01
- WF-DT-IMPORT-MAPPING-ERROR-02

Each frame must reference:

- UX-0004 flow ID.
- UX-0005 screen ID.
- Related component IDs from UX-0007 or UX-0011.
- Accessibility acceptance from UX-0009.
- Data or architecture dependency where behavior is not purely presentational.

4. Annotation Standard

Every frame includes an annotation panel containing:

- Screen purpose.
- Entry conditions.
- Primary user goal.
- Primary action.
- Secondary actions.
- Exit conditions.
- Required data.
- Loading behavior.
- Empty behavior.
- Error behavior.
- Offline behavior.
- Accessibility behavior.
- Analytics events, excluding private content.
- Open questions, if any.

Interaction annotations use:

- A — action.
- N — navigation.
- S — system response.
- AI — AI-generated or AI-assisted behavior.
- P — privacy or consent boundary.

- SYNC — synchronization behavior.
- ERR — error and recovery.
- A11Y — accessibility requirement.

5. Platform Canvases

5.1 Smartphone

Use representative compact and large smartphone widths. Layout is single-column unless a native overlay, sheet, camera, or system picker is shown.

Required shell:

- Top app bar or native navigation title.
- Main content region.
- Bottom navigation: Home, Transactions, Review, Search, Settings.
- Globally prominent Capture action.
- Safe-area and keyboard behavior.

5.2 Tablet

Use compact and expanded tablet widths.

Required patterns:

- Adaptive sidebar.
- List-detail or source-detail split view.
- Collapsible secondary pane where needed.
- Touch, pointer, keyboard, drag-and-drop, and rotation annotations.
- State preservation when layout changes.

5.3 Desktop

Use a resizable desktop window with minimum, default, and wide states.

Required patterns:

- Persistent sidebar.
- Global toolbar with Search, Import/Capture, and status access.
- Multi-pane work areas where they reduce context switching.
- Tables and batch actions for appropriate tasks.
- Keyboard shortcuts and focus order.
- Window and panel restoration behavior.

6. Global Shell Wireframes

Create the following shell frames for all three platforms:

6.1 Default Local Shell

Shows primary navigation, current destination, global actions, and status access.

6.2 Offline Shell

Local work remains available. A quiet offline indicator explains that changes are saved locally. Future optional synchronization must never be assumed or required.

6.3 Future Optional Sync-Delayed Shell

Future-design reference only. If optional synchronization is later implemented and enabled, show actionable delay status without blocking local work.

6.4 Security Alert Shell

Shows a persistent, privacy-safe alert for device trust, revocation, or identity risk.

6.5 Large Text / Accessibility Shell

Demonstrates reflow, visible focus, and non-color state communication.

7. Onboarding and Local Trust Frames

Required smartphone-first frames:

- Welcome and value proposition.
- Local encrypted storage explanation.
- Privacy summary.
- Initialize a local Butlerly workspace without creating an account.
- Restore or import local data when the user explicitly chooses to do so.
- Optional local device authentication for sensitive actions; it must not gate ordinary core use.
- Optional Apple synchronization explanation.
- External AI consent explanation.
- First useful task selection.
- Onboarding complete.

Required alternate states:

- Apple service unavailable.
- Identity import invalid.
- Authentication declined.
- Default single-device local use.
- Recovery path.

Tablet and desktop adaptations must preserve the same sequence but may use centered panels or split explanatory layouts.

8. Home Frames

8.1 Default Home

Required regions:

- Greeting or neutral context header.
- Capture and Import actions.
- Needs Attention summary.
- Recent records.
- Continue Recent Work when applicable.

- Local/offline or synchronization status only when meaningful.

8.2 Empty Home

Explains what Butlerly does and offers Capture, Import, or Sample Data without pressure.

8.3 Needs Attention Home

Prioritizes meaningful exceptions rather than engagement metrics.

8.4 Continue Recent Work

Shows task name, originating device, last activity time, Resume, and Dismiss. Sensitive details remain hidden by default.

8.5 Home Error and Recovery

Shows preserved local data and safe next actions.

9. Capture and Manual Entry Frames

Required smartphone frames:

- Capture launcher.
- Camera permission explanation.
- Camera view.
- Multi-page capture.
- Image-quality warning.
- Crop/rotate controls.
- Capture review.
- Processing locally.
- External processing consent when required.
- Extraction result.
- Low-confidence field review.
- Save success with Undo.
- Manual entry.

Required alternatives:

- Camera denied.
- No camera available.
- Import from Photos or Files.
- Offline capture.
- Processing failure with source preserved.
- Duplicate suspected.

Tablet wireframes show source and record side by side. Desktop supports file drop, file picker, scanner/import source, and multi-file staging.

10. Import Frames

Required desktop-first frames:

- Import source selection.
- File selection/drop zone.
- Batch staging.
- Column or field mapping.
- Validation summary.
- Duplicate detection.
- Processing progress.
- Exceptions summary.
- Import complete.
- Import history.

Required states:

- Unsupported file.
- Password-protected file.
- Partial parsing.
- Provider unavailable.
- Interrupted import.
- Safe resume.
- Cancel with preserved source.

Tablet supports smaller batch import and drag-and-drop. Smartphone supports simple file and photo import but not complex mapping unless unavoidable.

11. Inbox and Review Frames

11.1 Inbox Default

Contains actionable review items grouped by meaning, not technical source.

Required filters:

- All.
- Needs Review.
- Duplicates.
- Sync and Device.
- Security.

11.2 Review Card

Shows:

- What Butlerly found.
- Why it matters.
- Evidence.
- Confidence when material.
- One primary action.
- Edit, Dismiss, or Defer as appropriate.

11.3 Low-Confidence Extraction

Displays proposed value adjacent to source evidence.

11.4 Duplicate Comparison

Shows both records, matched fields, differences, recommendation, Keep Both, Merge, or Not a Duplicate.

11.5 Synchronization Conflict

Shows competing values, device names, timestamps, evidence, recommendation, and Choose/Edit/Defer.

11.6 Empty Inbox

States that nothing requires attention without gamification.

11.7 Bulk Review

Desktop-only or tablet-expanded pattern for explicit multi-selection. Consequential actions require preview and confirmation.

12. Records Frames

Required frames:

- Records list.
- Records table on desktop.
- Filter and sort panel.
- Search within Records.
- Record detail.
- Edit record.
- Source preview.
- Related records.
- Version history.
- Version comparison.
- Recently Deleted.
- Restore record.
- Permanent deletion confirmation.

Record detail hierarchy:

1. Record identity and status.
2. Key structured fields.
3. Source evidence.
4. AI explanation or provenance when relevant.
5. Related records.
6. Activity and version history.
7. Secondary and destructive actions.

Smartphone uses progressive disclosure. Tablet uses list-detail or source-detail split. Desktop may use list, detail, and source panes.

13. Search Frames

Required frames:

- Global search entry.
- Recent searches.
- Search suggestions.
- Results grouped by type.
- Filter panel.
- No results.
- Search unavailable for remote-only data while offline.
- Result detail preview.

Search must show whether results come from local data, connected sources, or AI interpretation. The user must be able to navigate results without using the Assistant.

14. Assistant Frames

Required frames:

- Assistant empty state with example capabilities.
- User question.
- Processing state.
- Completed answer.
- Evidence disclosure.
- Related records.
- Suggested next action.
- Action preview.
- External processing consent.
- Uncertainty or incomplete information.
- AI unavailable.
- Correction feedback.

Answer structure:

1. Direct answer.
2. Evidence or source summary.
3. Uncertainty.
4. Related records.
5. Recommended next action.

Streaming must not continuously disrupt screen-reader users. The completion state is announced once.

15. Settings Frames

Required settings map:

- Profile and identity.
- Privacy & Data.

- AI & Processing.
- My Devices — future optional cross-device capability, not P0.
- Synchronization — future optional capability, not P0.
- Notifications.
- Appearance.
- Accessibility.
- Import and Export.
- Backup and Recovery.
- About and Diagnostics.

Each settings frame must distinguish:

- Device-specific preference.
- Synchronized preference.
- Identity-wide policy.
- External-provider dependency.

16. Future Cross-Device and Pairing Frames

Required frames:

- My Devices list.
- Device detail.
- Add Device method selection.
- Apple synchronization setup.
- QR pairing display.
- QR scanner.
- Verification phrase.
- Approval request.
- Approval denied.
- Initial synchronization.
- Device ready.
- AirDrop/manual transfer package creation.
- Package import.
- Sync delayed.
- Device recovery needed.
- Revoke device.
- Lost device guidance.
- Revocation complete.

This section is a future-design reference. Cross-device trust, pairing, synchronization, and revocation are not required for the Finance V1 MVP and must never introduce a mandatory Butlerly account.

17. Privacy, Export, and Deletion Frames

Required frames:

- Privacy summary.

- Data location.
- External processing history.
- Consent management.
- Export scope selection.
- Export preview.
- Authentication.
- Export progress.
- Export complete.
- Delete record.
- Delete category or relationship.
- Delete all Butlerly data.
- Retention and recovery explanation.
- Permanent deletion complete.

Consequences, scope, and recoverability must be explicit before confirmation.

18. Error and Recovery Frame Library

Create reusable frame patterns for:

- Inline field error.
- Area banner.
- Full-screen unavailable state.
- Provider degraded.
- Offline.
- Permission denied.
- Authentication failed.
- Source preserved after processing failure.
- Sync delayed.
- Integrity validation failed.
- Recovery package invalid.
- Retry safe.
- Retry may duplicate and requires review.

Every error frame answers:

- What failed?
- What was preserved?
- Did data leave the device?
- What remains usable?
- What can the user do next?

19. Accessibility Wireframe Requirements

Every wireframe must annotate:

- Screen title and heading hierarchy.
- Reading order.
- Initial focus.

- Focus return after modal closure.
- Accessible names for icon-only actions.
- Large-text reflow.
- Touch target minimums.
- Keyboard path.
- Screen-reader announcements for dynamic states.
- Non-color meaning.
- Alternatives for drag, hover, camera, voice, QR, and gestures.
- Reduced-motion behavior.

At least one accessibility variant is required for each major flow.

20. Responsive and Adaptive Behavior

Each major destination must include a platform adaptation sheet showing:

- Smartphone compact.
- Smartphone large.
- Tablet compact.
- Tablet expanded.
- Desktop minimum width.
- Desktop default.
- Desktop wide.

The sheet must identify:

- Elements that persist.
- Elements that move.
- Elements that collapse.
- Elements that become a sheet, pane, or dialog.
- State that remains device-specific.
- State that synchronizes.

21. Prototype Linkage

Wireframes must support clickable prototypes for these P0 journeys:

1. First launch to first saved record.
2. Capture receipt and correct low-confidence data.
3. Import records and resolve exceptions.
4. Review a duplicate.
5. Search and inspect source evidence.
6. Ask the Assistant and review evidence.
7. Reserved for future optional cross-device continuity.
8. Work fully offline with local data preserved.
9. Reserved for future synchronization-conflict validation.
10. Reserved for future device-trust validation.
11. Export selected data.
12. Delete and restore a record.

22. Wireframe Review Gates

Gate 1 — Structural Review

Confirms information hierarchy, navigation, and screen coverage.

Gate 2 — Flow Review

Confirms entry, completion, cancellation, error, recovery, and cross-device continuity.

Gate 3 — Accessibility Review

Confirms equivalent outcomes and annotated accessibility behavior.

Gate 4 — Privacy and Security Review

Confirms consent, authentication, data location, device trust, export, deletion, and notification privacy.

Gate 5 — Engineering Review

Confirms feasibility, state models, data dependencies, platform adaptation, and measurable acceptance criteria.

Gate 6 — Usability Readiness

Confirms prototypes can test the intended hypotheses without visual polish masking structural issues.

23. Deliverable Package

The complete package includes:

- Master wireframe file.
- Cover and legend.
- Screen map.
- Flow map.
- Platform adaptation sheets.
- P0 frame set.
- P1 frame set where dependencies require early validation.
- State library.
- Accessibility variants.
- Clickable P0 prototypes.
- Annotation index.
- Open-issue register.
- Change log.

24. Version 1 Priority

P0:

- Global shells.
- Onboarding.
- Home.

- Capture and manual entry.
- Import.
- Inbox and review.
- Records and record detail.
- Search.
- Review and Search as primary Finance V1 destinations; Assistant remains optional and non-primary.
- My Devices — future optional cross-device capability, not P0.
- Offline local operation. Synchronization is deferred beyond P0.
- Privacy, export, deletion, and recovery.

P1:

- Advanced duplicate comparison.
- Rich version comparison.
- Detailed diagnostics.
- Recovery package management.
- Batch review enhancements.

P2:

- Family or delegated access.
- Android and Windows adaptations.
- Advanced selective synchronization.

25. Dependencies

UX-0010 depends on:

- UX-0001 — Butlerly Experience Blueprint.
- UX-0002 — Butlerly Version 1 User Journey Map.
- UX-0003 — Butlerly Information Architecture and Navigation Model.
- UX-0004 — Butlerly Core User Flows.
- UX-0005 — Butlerly Screen and State Inventory.
- UX-0006 — Butlerly AI Interaction and Explainability Specification.
- UX-0007 — Butlerly Design System Foundations.
- UX-0008 — Butlerly Cross-Device Experience and Continuity.
- UX-0009 — Butlerly Accessibility Standard.

UX-0010 governs:

- UX-0011 — Design Patterns and Component Library.
- UX-0012 — Usability Testing and UX Metrics.
- Clickable prototypes.
- High-fidelity UI design.
- Engineering UI stories and estimates.
- Design QA and release acceptance.

26. Acceptance Criteria

This specification is complete when:

- Every P0 screen and state has a required wireframe.
- Smartphone, tablet, and desktop adaptations are explicit.
- Normal, empty, loading, offline, degraded, error, success, and recovery states are covered.
- Privacy, export, deletion, local recovery, and offline behavior are represented. Cross-device synchronization, device trust, pairing, and merge behavior are deferred future-design references and are not required for P0.
- Accessibility annotations are included for every major flow.
- Wireframes can be assembled into the approved P0 Finance V1 prototypes; deferred sync/device-trust prototypes are excluded.
- Engineering can derive component, state, and data requirements without inventing product behavior.
- Usability researchers can test structure and comprehension before visual styling.
- Traceability exists from journey to flow to screen to wireframe to component and test.

BDS-0000 Conformance

All wireframes must implement BDS-0000 — Butler Design System v1.0 for semantic tokens, theme values, typography, spacing/radius, breakpoints, component/state behavior, accessibility, localization/original-data handling, and Flutter UI constraints. Finance V1 visible primary navigation is Home, Transactions, Review, Search, and Settings; Assistant is optional/non-primary. Any older wireframe labels using Inbox or Records as top-level destinations must be interpreted as Review or Transactions respectively, while legacy screen IDs may remain for traceability. Core operation requires no Butlerly account; device trust and synchronization are future/optional unless product scope explicitly enables them. UI labels follow the selected application language, while user-entered/imported/scanned/source data remains in its original language by default and original currency/amount remains authoritative.